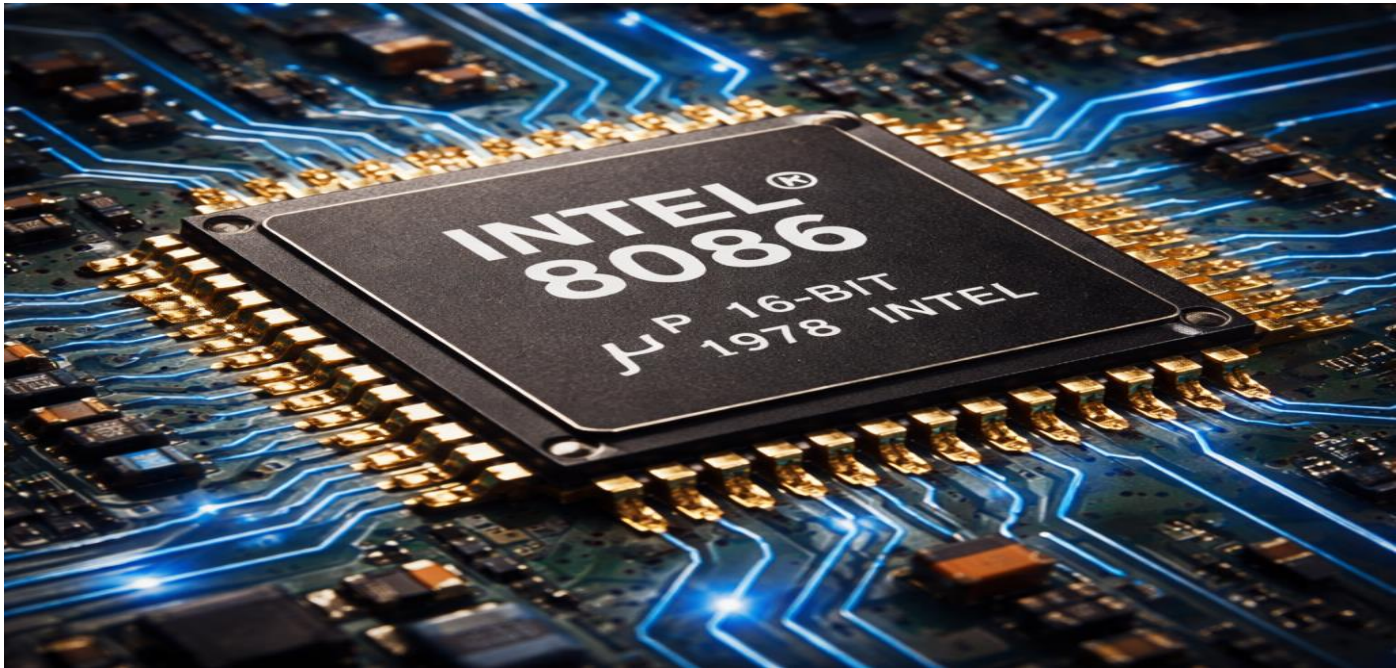


8086 family assembly language



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Debugger Commands

5. Register Command

This command allows us to examine or modify the constraint of registers.

Syntax:

- ***R*** [REGISTER NAME]

EX1 :

- **R**

```
-R
AX=0000 BX=0000 CX=0000 DX=0000 SP=00FD BP=0000 SI=0000 DI=0000
DS=073F ES=073F SS=073F CS=073F IP=0100  NV UP EI PL NZ NA PO NC
073F:0100 0000      ADD     [BX+SI],AL      DS:0000=CD
-
```

Examples:

EX2:

```
-R BX
BX 0000
:558F
-R
AX=0000 BX=558F CX=0000 DX=0000 SP=00FD BP=0000 SI=0000 DI=0000
DS=073F ES=073F SS=073F CS=073F IP=0100  NU UP EI PL NZ NA PO NC
073F:0100 0000          ADD     [BX+SI],AL          DS:558F=00
-
```

EX3:

```
-R CX
CX 0000
:F3F
-R
AX=0000 BX=558F CX=0F3F DX=0000 SP=00FD BP=0000 SI=0000 DI=0000
DS=073F ES=073F SS=073F CS=073F IP=0100  NU UP EI PL NZ NA PO NC
073F:0100 0000          ADD     [BX+SI],AL          DS:558F=00
-
```

Examples:

EX4: Modify the content of DS register to (F900).

```
-R DS
DS 073F
:F900
-R
AX=0000 BX=558F CX=0F3F DX=0000 SP=00FD BP=0000 SI=0000 DI=0000
DS=F900 ES=073F SS=073F CS=073F IP=0100 NU UP EI PL NZ NA PO NC
073F:0100 0000          ADD     [BX+SI],AL          DS:558F=00
-
```

EX5: Store the values: 55 66 77 88 99 in a block of memory having 5 locations, starting address is 4000:0100

```
-R DS
DS F900
:4000
-R
AX=0000 BX=558F CX=0F3F DX=0000 SP=00FD BP=0000 SI=0000 DI=0000
DS=4000 ES=073F SS=073F CS=073F IP=0100 NU UP EI PL NZ NA PO NC
073F:0100 0000          ADD     [BX+SI],AL          DS:558F=00
-F DS:0100 0104 55 66 77 88 99
-D DS:0100 0104
4000:0100 55 66 77 88 99          Ufw..
-
```

Debugger Commands:

6. Assemble Command

الأمر A يسمح بإدخال برنامج بلغة (assembly) إلى الذاكرة. ويحدد عنوان (offset address) الذي سيتم كتابة التعليمات به ويتعامل دائماً مع (code segment) لأن إيعازات البرنامج تخزن داخل (code segment).

Syntax:

- A [Address]

Example:

```
-R
AX=0000 BX=558F CX=0F3F DX=0000 SP=00FD BP=0000 SI=0000 DI=0000
DS=4000 ES=073F SS=073F CS=073F IP=0100  NU UP EI PL NZ NA PO NC
073F:0100 0000          ADD     [BX+SI],AL          DS:558F=00
-A 0100
073F:0100 Mov AX,1000
073F:0103 Mov SI,0100
073F:0106 Add AX,SI
073F:0108 Mov DL,[SI]
073F:010A
```

Debugger Commands:

7. Trace Command

يستخدم هذا الأمر لتتبع تنفيذ إيعازات البرنامج خطوة خطوة ويبين ماذا يحدث لقيم (REGESTERS).

Syntax:

- T [=Offset Address]

لاحظ تنفيذ البرنامج السابق:

```
0F-R
- AX=0000 BX=558F CX=0F3F DX=0000 SP=00FD BP=0000 SI=0000 DI=0000
- DS=4000 ES=073F SS=073F CS=073F IP=0100 NU UP EI PL NZ NA PO NC
073F:0100 B80010 MOV AX,1000
ID -T=100
AX=1000 BX=558F CX=0F3F DX=0000 SP=00FD BP=0000 SI=0000 DI=0000
DS=4000 ES=073F SS=073F CS=073F IP=0103 NU UP EI PL NZ NA PO NC
073F:0103 BE0001 MOV SI,0100
-T
AX=1000 BX=558F CX=0F3F DX=0000 SP=00FD BP=0000 SI=0100 DI=0000
DS=4000 ES=073F SS=073F CS=073F IP=0106 NU UP EI PL NZ NA PO NC
073F:0106 01F0 ADD AX,SI
-T
AX=1100 BX=558F CX=0F3F DX=0000 SP=00FD BP=0000 SI=0100 DI=0000
DS=4000 ES=073F SS=073F CS=073F IP=0108 NU UP EI PL NZ NA PE NC
073F:0108 8A14 MOV DL,[SI] DS:0100=55
-T
AX=1100 BX=558F CX=0F3F DX=0055 SP=00FD BP=0000 SI=0100 DI=0000
DS=4000 ES=073F SS=073F CS=073F IP=010A NU UP EI PL NZ NA PE NC
073F:010A 0000 ADD [BX+SI],AL DS:568F=00
-
```

H.W

You have the following two blocks of memory, do the following:

- 1- Fill the block1 and block2 with value as shown in the figure.
- 2- Display the content of memory location of two blocks.
- 3- Exchange the contents of two block to with each other.
- 4- Display the content of the blocks.

Block 1

4000:2005	5F
2006	4A
2007	55
2008	88

Block 2

2000:F9AA	66
F9AB	77
F9AC	88
F9AD	B2